UTSAV CHAUDHARY

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Field Service Engineer supporting warehouse **robotics** on live customer floors. Recover **bot errors** in real time, repair hardware in the **maintenance cage**, replace parts, and close **JIRA** and **ServiceDesk** tickets with customers and remote support.

Technical Skills

Robotics & Field Support: AMR, AGV, warehouse automation, real-time bot error recovery, GM Tower, maintenance cage, barcode management, fulfillment and distribution sites

Hardware Repair: Troubleshooting, parts replacement, reassembly, root cause analysis, escalation, Linux, SSH, firmware, soldering, oscilloscope, multimeter, bench debugging

Embedded, IoT & Robotics: Raspberry Pi, Arduino, ATmega2560, embedded C/C++, Python, IR, ultrasonic, thermal and line-follow sensors, IMU, camera modules, motor drivers, actuators, GPIO, I2C, SPI, UART, MQTT, sensor integration, prototyping, firmware flashing, edge AI, YOLOv8

Tickets & Coordination: JIRA, ServiceDesk, database updates, issue documentation, customer communication, remote hardware support, nights, weekends, shift coverage

Experience

Field Service Engineer, Hybrid, **MidAmerican Energy**, TX — Feb 2025 - Present

- Recovered **bot errors** in real time across a **[300+]** unit **AMR/AGV** fleet, clearing **[40+]** faults per shift without pulling robots off the floor and holding **[99%]** fleet availability through peak waves.
- Repaired bots end to end in the **maintenance cage** (diagnose, replace parts, reassemble, release), returning **[12+]** units per week and cutting average turnaround from **[48 hrs]** to **[12 hrs]**.
- Logged faults in **GM Tower** and drove **[20+] JIRA/ServiceDesk** tickets per week to closure with remote support, sustaining **[95%]** on-time ticket closure against SLA.
- Escalated recurring faults to the maintenance manager with repair history and **root cause** data, reducing repeat failures on problem aisles by **[30%]** quarter over quarter.

Field Support Engineer, On-Site, **Chevron**, GA — Jan 2024 - Jan 2025

- Repaired **bot** hardware during live warehouse shifts, reassembling and revalidating **[98%]** of units for same-shift return to the floor instead of overnight hold.
- Recovered floor errors and owned escalation of repeats to maintenance, cutting mean time to repair from **[6 hrs]** to **[2 hrs]** across the site.
- Audited **barcode** and database records on every parts swap, keeping **100%** serial-to-unit traceability on the fleet for **[12]** months with zero record mismatches at audit.

Projects

Firebird V Autonomous Robot on ATmega2560

- Wrote **embedded C/C++** firmware on the **ATmega2560** for the **Firebird V** robot, covering **motor control**, **encoder** feedback, and **closed-loop** steering so the robot held its line and cleared the course without losing position.
- Owned the **hardware** side as well: wired **IR sensors**, **IMU**, and **serial** links, brought each board up from bare silicon, and debugged faults at the board level until the build ran clean.

Smart Fire Hazard Detection Using Thermal AI and Edge Computing

- Built a **Raspberry Pi** monitor with **RGB** and thermal cameras, and ran **YOLOv8** on the edge for fire and smoke detection at **85%** accuracy without sending every frame off the device, so the alert stayed fast.
- Used a cloud-hosted **Faster R-CNN** with **MQTT** for thermal anomaly detection, reaching **95%** accuracy.

Automated Ambu Pump (Open-Source COVID-19 Response)

- Built an **automated Ambu pump** with an open-source team during **COVID-19** for **hands-free ventilation**.
- Designed the mechanical drive with **3D-printed** gears and enclosure, **DC motor** and **servo** actuation, and tuned the squeeze cycle for a steady breaths-per-minute rate through hours of continuous running.
- Wrote the **Arduino** control code in **embedded C++** for adjustable rate and depth, and validated squeeze pressure and timing on the bench with the team before each unit was shipped.

Education

State University of New York at Binghamton — Aug 2022 to Dec 2023
Master of Science in Computer Science — 3.4/4.00 GPA

Universal College of Engineering, Mumbai — July 2017 to June 2021
Bachelor of Engineering in Computer Engineering — CGPA 8.09/10